

DATUM LINE

AI × BIO · 29 SEPT

# “Do my cells look normal?”

Drop an image.

Get an answer against every other lab.



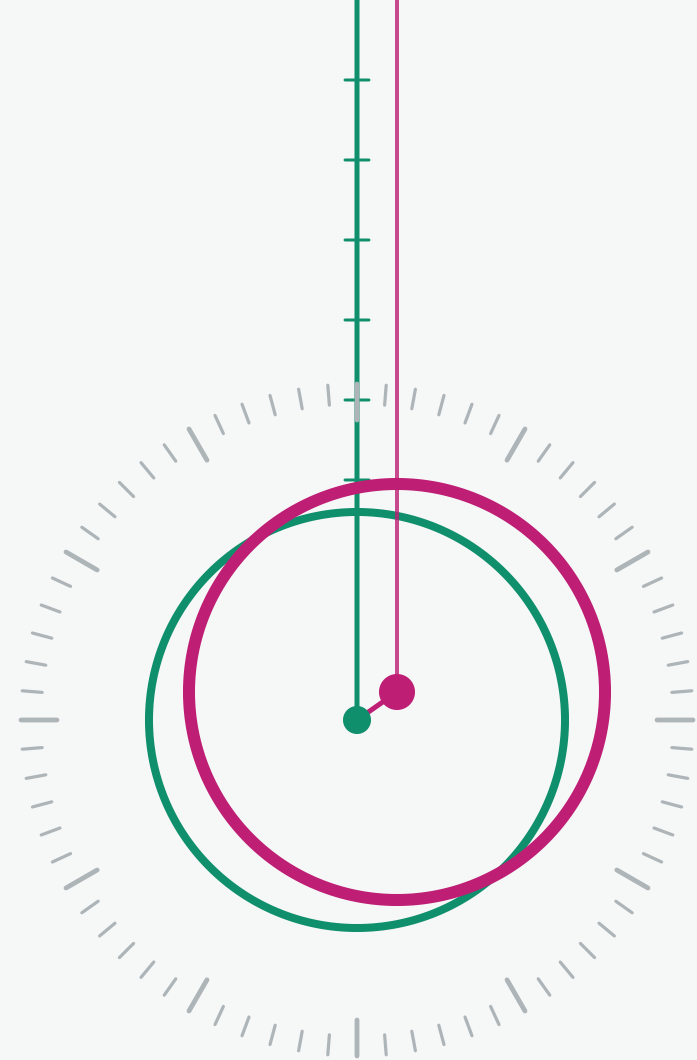
control image, any microscope  
no compound, no target, no result

ask

Labs ask for free.

Companies that buy cell data pay for the certificate.

Kat Titterton · San Francisco



# One lab, or six?



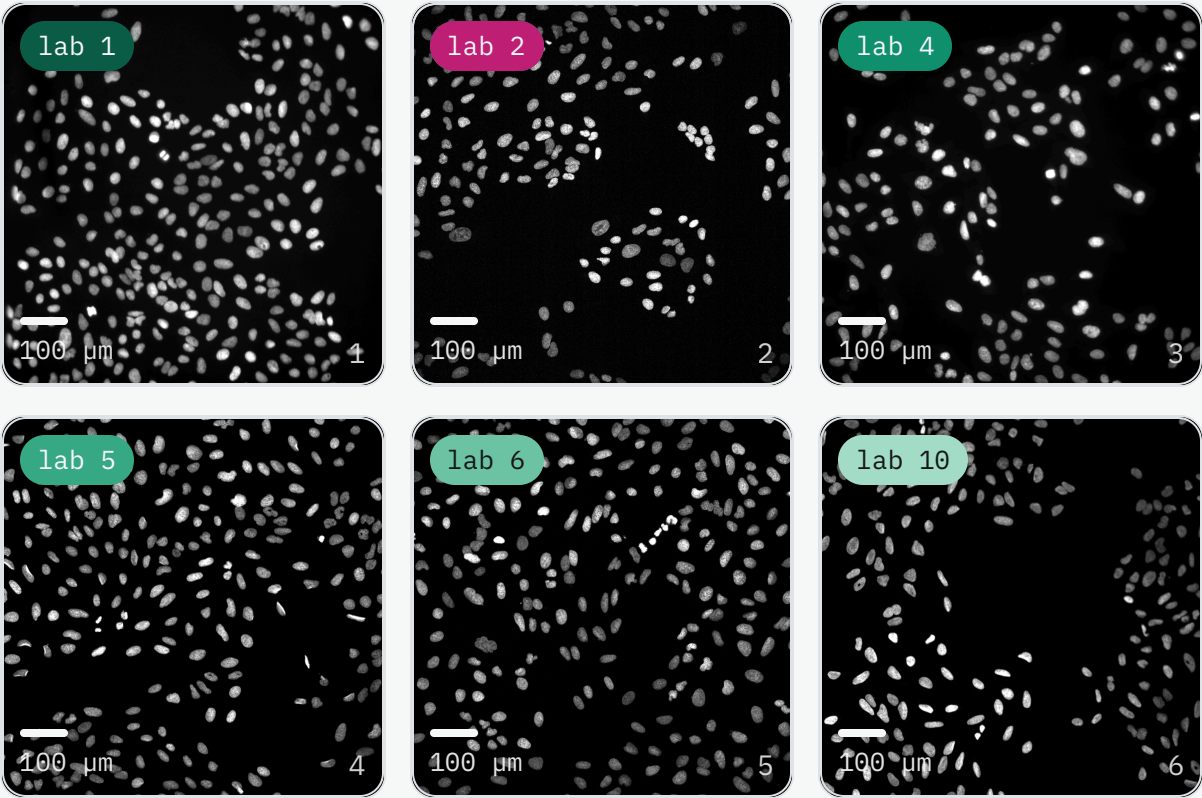
Cell Painting five dyes, 3,773 numbers per well

-  same line ✓
-  same protocol ✓
-  no drug ✓

Fig. 1 93,228 untreated wells, 11 labs, plates held out. JUMP [7]

[click → the answer](#)

# Could you tell? A classifier can.



Cell Painting five dyes, 3,773 numbers per well

99.9% CLASSIFIER

9% CHANCE, 1 IN 11 LABS

78 wells wrong, of 93,228

Normalise each plate to its own controls: 99.9% becomes 24%. A3

Fig. 1 93,228 untreated wells, 11 labs, plates held out. JUMP [7]

click → follow one well

# One well, all the way out.

ONE NUCLEUS



192  $\mu\text{m}^2$  · 0.59

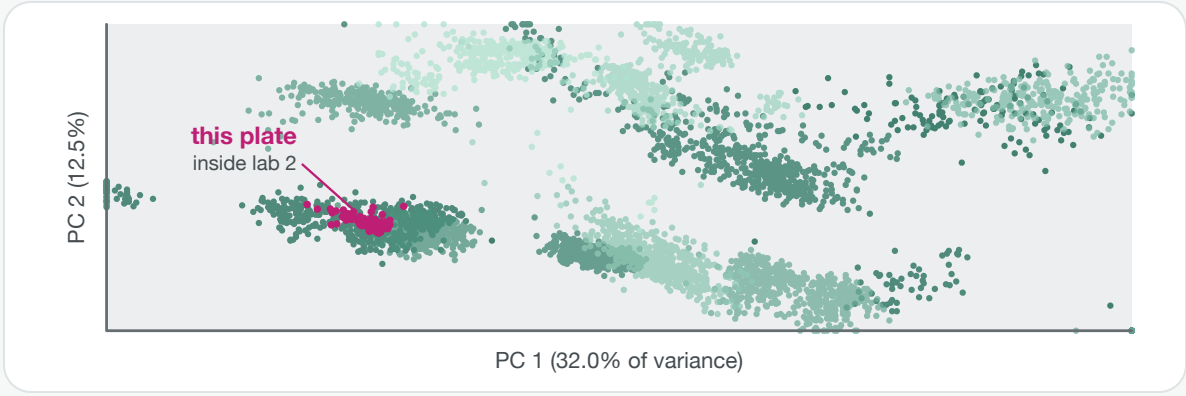
ITS WELL



plate 1053600681

3,773  
numbers  
→  
one point

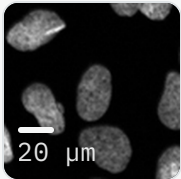
WHERE IT LANDS



93,228 wells, 11 labs

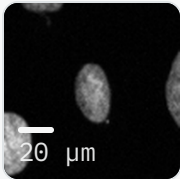
other labs

lab 5



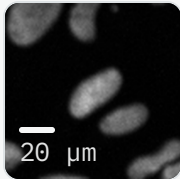
195  $\mu\text{m}^2$  · 0.94

lab 6



213  $\mu\text{m}^2$  · 0.50

lab 10



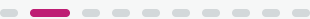
207  $\mu\text{m}^2$  · 0.29

AREA     ≈ ≈  
TEXTURE    ≠ ≠

One lab's normal is not another's. The difference is shape, not count.

Fig. 1 93,228 untreated wells, 11 labs, plates held out. JUMP [7]

click → start over



# Today: search, ask, guess.

## WHY NOW

Since 2023 the field's reference is public [7].

Since 2025 cell data carries regulatory weight [9]. Since 2025 cell data carries regulatory weight [9].

IRREPRODUCIBLE PRECLINICAL RESEARCH · US · PER YEAR [6]

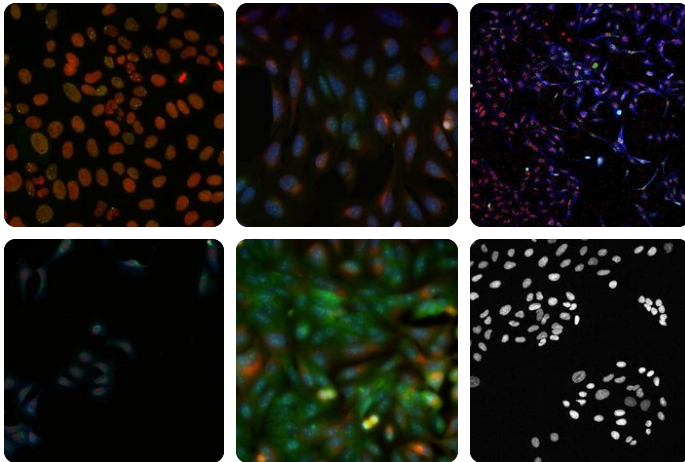
\$28B

36% ≈ \$10B · reagents & references

64% · design, analysis, protocols

### 1 search

Q do my U2OS cells look normal



### 2 ask around

To: whoever ran this line last

Subject: does this look right to you?

Attached: screenshot.png

### 3 guess

\$ RUN IT ANYWAY · TOO FAST

**hundreds of  
thousands**

of R&D before the SOP was fixed

🕒 HOLD · TOO SLOW

**months**

developing cells that were already good enough

[6] Freedman et al., *PLoS Biol*, 2015. Search results: real U2OS images from six papers.



[macrofluidic.me/resume](https://macrofluidic.me/resume)

TEAM

One founder today.  
Looking for a technical co-founder.  
Collaborating labs: insitro, Axiom Bio, Soley.

# Katherine Titterton

Five labs, and the same **unanswered question** at every bench. It made the science slow and wrong.

● 2023–2026

Axiom

SAN FRANCISCO

founding MTS • hepatocytes at scale

● 2022–2023

insitro

SOUTH SAN FRANCISCO

neuroscience assays • productionization

● 2021–2022

MIT

SYNTHETIC NEUROBIOLOGY

expansion microscopy

● 2018–2021

abbvie

NEUROSCIENCE

tau • neurons

● 2017–2018

 HARVARD MEDICAL SCHOOL

Xenopus oocytes • GABA<sub>A</sub> receptors

● 2017

COLGATE

BA

neuroscience + math biology

MY WORK

- 1

Cell Painting in primary human hepatocytes  
*Cell Systems* 2026 • Axiom Bio
- 2

Counting cells predicts bioactivity benchmarks  
*Nature Communications* 2026 • Axiom Bio
- 3

A neuronal tau aggregation assay  
*J Biol Chem* 2025 • AbbVie
- 4

Multiplexed expansion revealing  
*Nature Communications* 2024 • MIT
- 5

Neurosteroid photolabeling of GABA<sub>A</sub> receptors  
*Eur J Med Chem* 2019 • MGH • Harvard Medical School

PUBLISHED WITH

Broad Institute • MIT • AbbVie • EMBL-EBI • HESI • Bayer  
• University of Cambridge • Uppsala University



# Where you sit predicts what you measure.



INSIDE LABS

$$\rho = -0.71$$

the further from your baseline,  
the further from your own answers

lab 2 against its own June answer

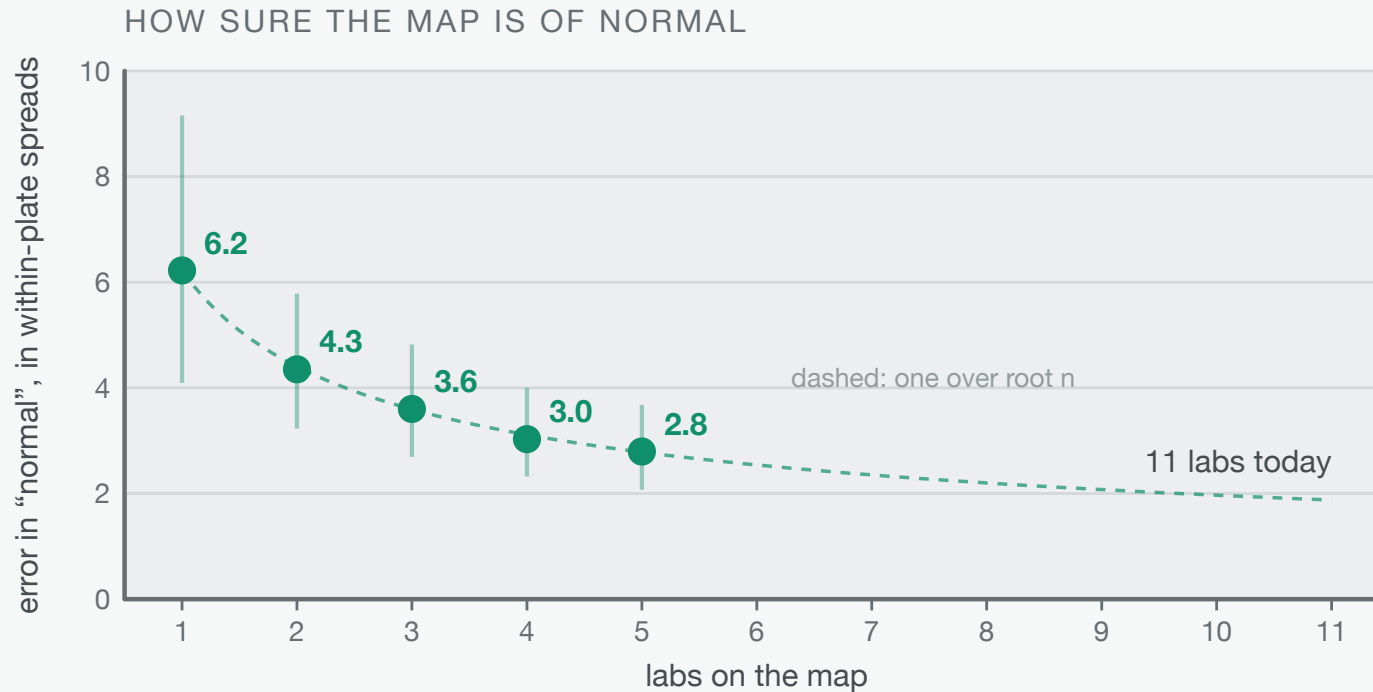
**0.85** before the flag

**0.67** after it

Fig. 3 One point per batch: 118 batches, 11 labs. JUMP [7]. More: A8, A13, A14

click → start over

# Every lab that joins **sharpens the map.**



## THE MAP'S ERROR

**6.2 → 2.8**

one lab: normal is 6.2 off.

Five labs: 2.8.

Smaller error, smaller oddities seen.

## your own baseline

**0.9 → 0.4**

from one plate to eight, then flat

COMMUNITY **open**

map, box, code and public reference

DATA ACCESS **paid**

same data, structured and comparable; pre-cleaned on request

THE BENCHMARK **co-built**

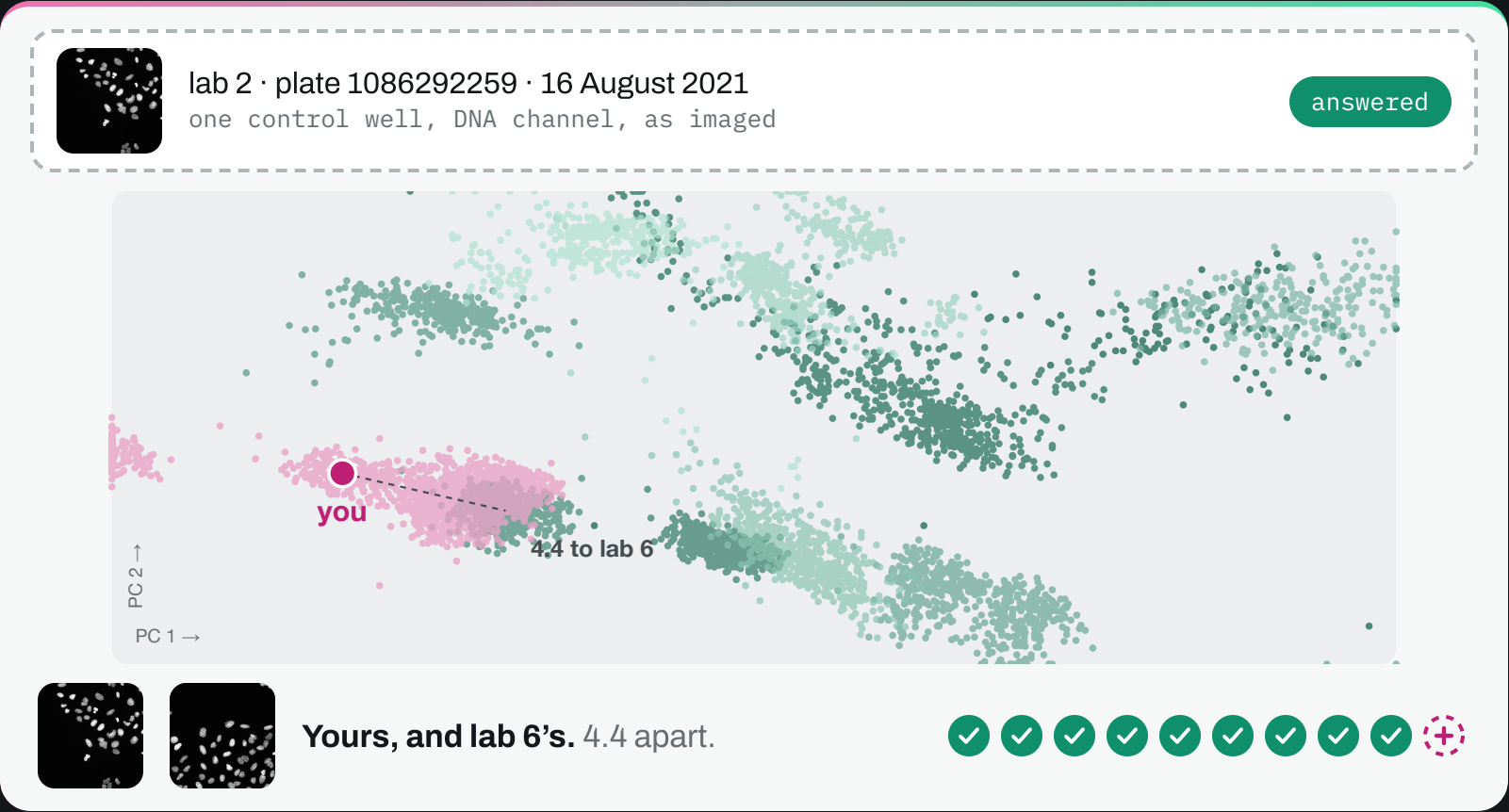
planted failures; every model graded, segmentation to bioactivity

Fig. 4 Error: distance between two maps built from different labs. 1,871 plates, 11 labs. JUMP [7]. More: A14

Prices and terms proposed. A15

# Drop an image. **Get an answer.**

Where your cells sit, whose they look like, and their cells next to yours.



## The answer

LAB 2 · 16 AUGUST 2021



YOUR CELLS  
U2OS · 25 plates

WHERE YOU SIT  
**in distribution**

2.8 from your baseline · 4.4 to lab 6, the nearest

YOUR OWN CHECK  
188 cells per well · in range

YOUR REFERENCE  
1,652 plates · 10 labs

YOUR DATA, MADE COMPARABLE  
agreement with other labs **0.70 → 0.77**

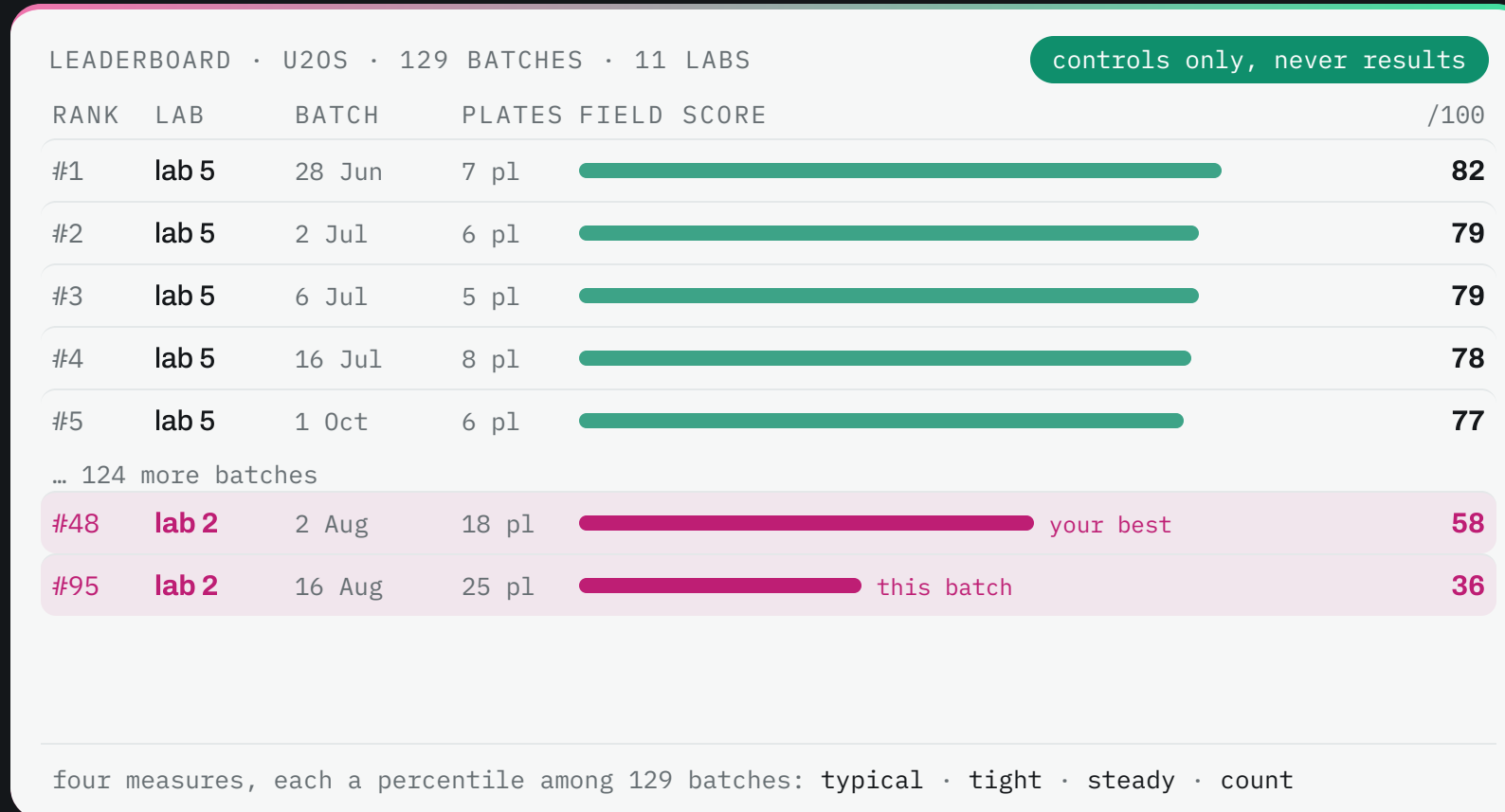
This page is the certificate. Held by nobody who sells you cells.

Fig. 5 Lab 2 on 16 Aug, replayed against the published field. JUMP [7]. More: A11

[click → the leaderboard](#)

# A game you can win.

Every batch ranked on its controls alone: **how typical, how tight, how steady.**  
Never on results.



click → start over

## Your standing

LAB 2 · 16 AUG

#95 of 129 batches · score 36

YOUR BEST

#48, 2 Aug · score 58

YOUR STREAK

9 batches in a row · the field's longest: 33, lab 5

TO CLIMB

steady: 2.9 against 1.2 for the field

## THE LAB TO BEAT

lab 5 · median score 70 over 33 batches

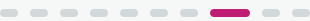
```
closest to the centre: lab 3 .
tightest plates: lab 5 . longest
streak: lab 5
```

# Five questions, five tools.

Each answers a different question about your cells.  
Use all of them. One did not exist.

| TOOL                | THE QUESTION IT ANSWERS                         | HOW                                                                     | WHAT IT DOES NOT DO                                                                    |
|---------------------|-------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| STR authentication  | Is it the line the label says?                  | DNA fingerprint against a database; standard for identity               | nothing about how the cells behave                                                     |
| Vendor certificate  | Was it fine when it shipped?                    | identity, sterility and viability at the bank                           | nothing after it leaves the bank                                                       |
| Plate controls      | Did this plate work?                            | control wells on every plate; catches a bad well or plate               | cannot see a whole lab drift, because the controls drift with it                       |
| Phenomics platforms | Is my own data consistent?                      | batch correction inside one organisation's data                         | removes lab differences by design; nothing across labs                                 |
| Datum Line          | Where do my cells sit, against every other lab? | one map of untreated wells from every lab; a place, a distance, a frame | cannot tell which line it is (that is STR); fluorescence today, brightfield is the bet |

Partners, not rivals: an identity check and a behaviour check belong on the same certificate. More: A7, A3



# Eighteen months, four milestones.

**\$1.0M**

pre-seed, eighteen months. Built line by line on A21.

Each milestone is something a partner lab can see on the map.

# 1

## Three labs on the map

collaborating labs: insitro, Axiom Bio, Soley. U2OS controls, live answers every batch

# 2

## Model that reads any image

the map from the picture itself,  
brightfield included: the bet on  
A10

# 3

## Benchmark, with two partners

two companies contribute failures and get their methods graded. A16

4

## First paid certificate

on a dataset a company bought  
and could not align

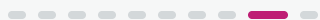
TODAY

One founder. Eleven labs of public data, one cell line, fluorescence.  
Every number in this deck.

MONTH 18

- Three partner labs live, a model that reads any image, the benchmark with two partners, and a first paid certificate.

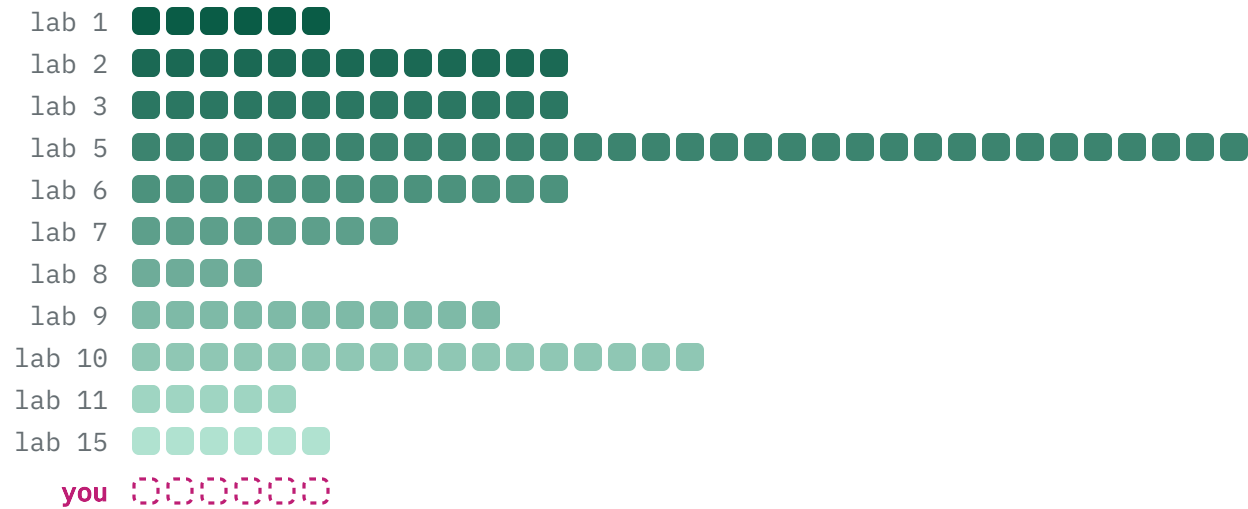
Proposed. The proof of concept sets the size.



# Every plate you image, placed on the map.

One map for people and for agents, held by nobody who sells you cells.  
Everyone learns and grows together.

FIG. 7 • ALREADY ON THE MAP 129 batches • 11 labs • JUMP [7]



1,871 plates, one map. Next row is yours.

- NOW

U2OS, fluorescence
- NEXT

more lines, brightfield
- THEN

the baseline a field shares

WHAT I NEED

-  **Images** U2OS controls
-  **People** technical co-founder; ML better than me
-  **Bench** 3-lab proof, then the set
-  **Pre-seed** to run it
-  **Partners** two, for the benchmark

Have images of U2OS cells? Talk to me.

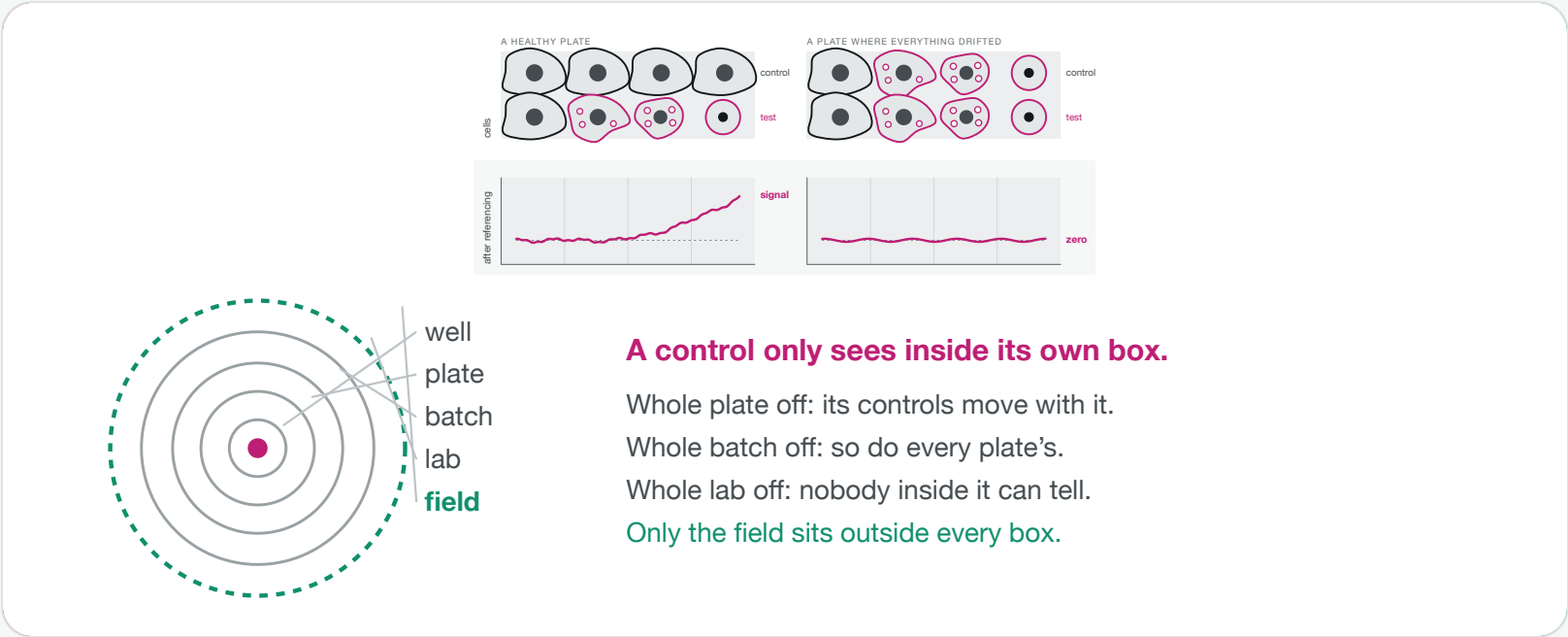
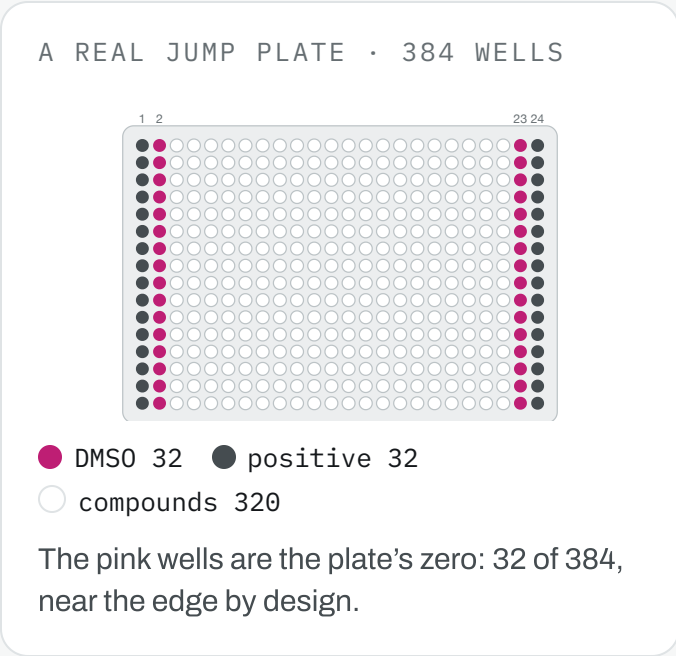
# The rest, if you ask.

Find the question, jump to the board. Every number traces to a file in results/ or a numbered reference; pending work says pending.

|     |                                                                        |     |                                                                                     |
|-----|------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------|
| A1  | <b>Why not trust the plate’s controls?</b><br>There is no control well | A12 | <b>What moved?</b><br>Stain intensity, not cell count                               |
| A2  | <b>Show me the math.</b><br>Six operations on one embedding            | A13 | <b>How does one lab move over time?</b><br>The lab’s check and the field’s view     |
| A3  | <b>Why not batch-correct it away?</b><br>What JUMP records             | A14 | <b>What does the field’s frame buy?</b><br>Agreement between labs                   |
| A4  | <b>Is the model the moat?</b><br>Counting cells often wins             | A15 | <b>Who pays?</b><br>Three buyers, one certificate                                   |
| A5  | <b>Why not just count cells?</b><br>Two ways to ask one question       | A16 | <b>What would the money build?</b><br>The ugliest dataset, on purpose               |
| A6  | <b>How often do I re-certify?</b><br>The line is not expired           | A17 | <b>Is the leaderboard fair?</b><br>Controls only, never results                     |
| A7  | <b>Isn’t this STR?</b><br>STR says who, images say how                 | A18 | <b>What is private?</b><br>Control wells leave; nothing else does                   |
| A8  | <b>Is the assay itself working?</b><br>A known answer                  | A19 | <b>What is the machine learning?</b><br>Image in; place, distance, move out         |
| A9  | <b>Does it change the result?</b><br>Same drug, different answer       | A20 | <b>What about biosecurity?</b><br>Cells cannot lie about where they came from       |
| A10 | <b>Does it work without stains?</b><br>Bet on brightfield              | A21 | <b>How much, and what for?</b><br>About \$1M for eighteen months, no lab of our own |

# There is no control well.

Every result is test minus control, on the same plate. But a whole plate can be off, or a whole batch, or a whole lab. A control only sees inside its own box.



$$x_{test} = \mu + s + \tau(s) \quad x_{ctrl} = \mu + s \quad \Delta = \tau(s)$$

$s$ , the plate's state, cancels from the number. Never from the cells.

So a control only sees inside its own box. If the plate is sick, so is the control. Then how would anyone know?

**Fig. A1** Where a control sits, and what it can see. Left, a real 384-well JUMP compound plate (lab 2, 1053601756): 32 DMSO, 32 positive, 320 compound wells. Right, schematic cells and traces, and the nesting of well, plate, batch, lab and field. No measurements shown.

# Six operations on one embedding.

Every well becomes a vector  $e \in \mathbb{R}^{3773}$ , z-scored across the field. Everything on the previous slides is one of these, and each one is a difference of means or a held-out classifier, so a lab can check it.

## 1 · EMBEDDING

**[1]**

$$e = \varphi(\text{image}); \varphi \in \{\text{CellProfiler}, \text{CNN}, \text{ViT}\}$$

Which  $\varphi$  barely matters: the three performed similarly on hepatocytes. The extractor is not the asset.

## 2 · LAB CLASSIFIER

**99.9%**

$$f: e \rightarrow \text{lab}, \text{ held out by plate}$$

Logistic regression, L2, five folds. Balanced accuracy, chance 1/11. Permuted labels 9.3%. Without the 1536-well labs: 100.0%.

## 3 · DISTANCE, THE CERTIFICATE

**[e6]**

$$d = \| E[e \mid \text{plate}] - E[e \mid \text{field}, c] \|_{\sigma}$$

Difference of two means in units of the field's spread, conditioned on c. Its direction is what moved;  $-d$  is the way back.

## 4 · VARIANCE DECOMPOSITION

**[e5]**

$$\text{Var}(e) = V_{\text{model}} + V_{\text{lab}} + V_{\text{batch}} + V_{\text{plate}} + V_{\text{well}}$$

Nested, so shares sum to one. Microscope model separates from lab because models are shared across labs.

## 5 · REGRESS VARIABLES OUT

**24%**

$$e_{\perp} = e - \hat{E}[e \mid c], \text{ then re-run } f$$

Remove only what you know is instrument, keep what may be biology. Remove each plate's own control mean and f falls to 24%.

## 6 · MORPHOLOGY CHANGE, TWO WAYS

 **$\delta$** 

$$\begin{aligned} \text{DISTANCE } \delta &= f(\|e - E[e \mid \text{vehicle}]\|), f \text{ bounded} \\ \text{CLASSIFIER } g: e' &\rightarrow \text{label}, e' \text{ aligned to the reference before comparison} \end{aligned}$$

Same vector, two questions. Discovery wants invariance to the lab: align first. QC wants the lab coordinate: never align it away.

results/e1, e1b, e5, e6 · [1] Ewald et al. 2026

# What JUMP records. What nobody does.

Normal is a conditional expectation,  $E[x \text{ / every variable below}]$ . Even JUMP, built to be comparable, records only the instrument side.

## RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

## NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze-thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity

## NORMALISE THE LAB AWAY AND THE LAB DISAPPEARS

- 99.9% raw features  
balanced accuracy naming the lab
  - 24% plate-centred  
own control mean removed; chance 9%
  - 19% plate-centred and scaled  
own control spread removed too
- 29,943 wells, 16 per plate, plates held out. results/e1-redteam.

Centre each plate on its own controls, the standard first step, and the lab is named 24% of the time. It removes most of the coordinate a certificate needs.

Batch correction is built to remove this signal: Arevalo J, et al., Nature Communications, 2024. Variables: JUMP datasets metadata.

# And the benchmarks are so weak that **counting cells** often wins.

If a naive baseline matches your model, the benchmark is the problem. The bottleneck was never the model. It is what you are comparing against.

# Two ways to ask the same question.

Why now: an image classifier can answer this, and until recently nothing could. A count moves only when cells die or divide; an embedding of the whole well moves when anything changes.

SAME CELL LINE, SAME PROTOCOL



|            |      |      |           |
|------------|------|------|-----------|
| cell count | 236  | 224  | same      |
| shape      | 0.94 | 0.50 | different |

METHOD ONE

Does this run belong with the past runs?

$$g: e' \rightarrow \text{belongs} / \text{does not}, \quad e' = \text{align}(e)$$

Align first, so the answer is the cells and not the day.

Standard alignment; the tuning is ours.

METHOD TWO

How far out, and which way back?

$$\delta = f(\|e - E[e \mid \text{vehicle}]\|), \quad f \text{ bounded}$$

Bounded, so one wild well cannot swamp the score.

Distance published with the Broad [1]; transform unpublished.

Both methods need the same thing to work: somebody else’s images. It takes a reference to know which.

**Fig. A5** Same cell line, two untreated wells, and the two readouts. Left, DNA channel, JUMP cpg0016, CC0; nuclei by Otsu segmentation; detail = high-frequency energy over total, inside each nucleus. Right, the two readouts: a classifier on aligned embeddings, and a bounded distance from the vehicle centre. Alignment and distance follow published practice [1]; the specific transform and tuning are not disclosed here. [1] Ewald JD, Titterton KL, et al. *Cell Systems*, 2026.

# The line is not expired. Your baseline is.

Two plates of one lab sit further apart the longer the gap between them. In lab 2 the distance is still rising after thirteen weeks.



IS THE CELL LINE RUINED?

Not known from profiles alone. The data says the lab moved and when. It does not say why.

SO WHY RE-CERTIFY?

Because your certificate is a comparison made on one day, and the baseline it was made against goes stale within weeks.

HOW SOLID IS THE NUMBER?

One lab, 219 plates, 13 batches, June to October 2021. 0.63 in the same batch, 4.7 over 13 weeks. Other labs are not yet measured.

Median distance between plate centroids, in within-plate spreads, with the interquartile range. results/e1b-drift.

# STR says who. Images say how.

Not a competitor: the other half of the same certificate.

## STR AUTHENTICATION



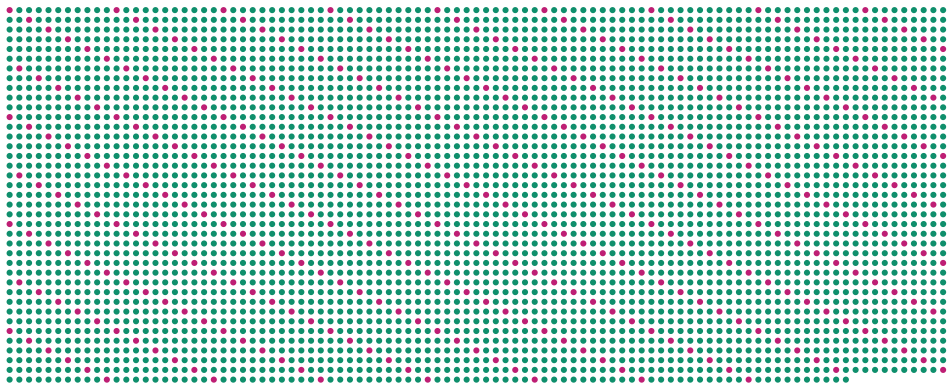
**17**  
loci

**once**  
per line, from  
a mailed card

**who**  
which line it  
is

The standard for identity. Two cultures with the same fingerprint can behave differently.

## DATUM LINE



**3,773**  
numbers per well

**every batch**  
from images already taken

**how**  
the way it behaves, against  
every lab

Behaviour, placed on the field's map. Cannot say which line it is.

identity · STR

+

behaviour · Datum Line

=



one certificate.

Partner, not rival: the STR provider is the natural first integration.

STR: ATCC cell authentication service, 17 loci plus amelogenin. Datum Line: 93,228 untreated wells, 1,872 plates, 11 labs, JUMP [7].

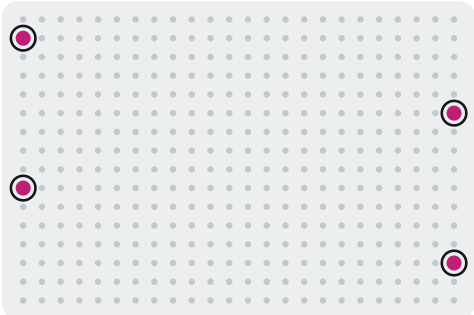
# A known answer, run in every lab.

Untreated wells certify where a lab sits. Positive controls certify that the assay answers at all. The field already runs the same eight.



## WHERE ON THE PLATE DOES AMG900 SHOW?

a batch that came out  
plate C13443aW



○ where the plate map puts it

a batch that failed  
plate A12069aW



● wells that look like it

### RUNS NOW ON JUMP

Every compound plate carries eight positive controls. In 125 of 129 batches the known answer came out.

### WHAT THE FIELD CAUGHT

Lab 3, 4 consecutive batches, 65 plates: the wells the plate map labels as positive controls show no effect, and no other well on the plate does.

### TWO CHECKS, TWO FAILURES

The untreated-well certificate flagged 1 of these 4 batches. A plate can sit in the right place and still not answer.

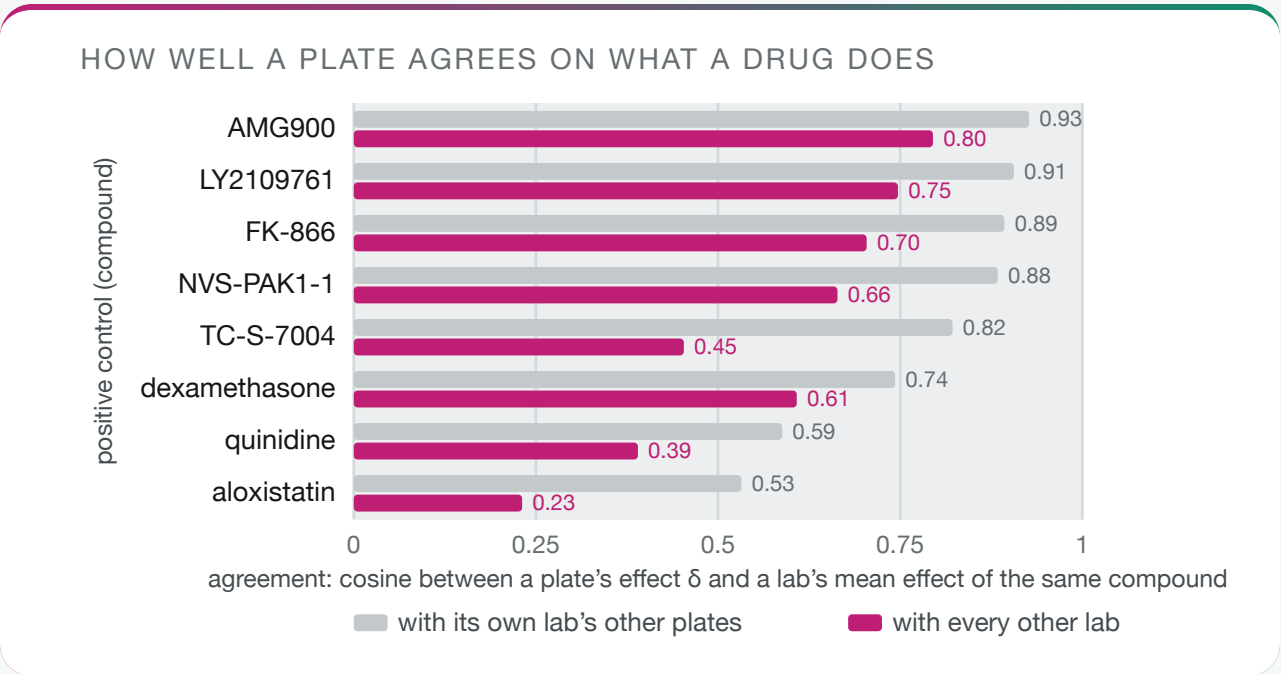
# Same drug. Different answer.

Labs running the same assay slightly differently rank the same compounds differently.  
A hit list made in one lab is not the hit list in another.

WHAT A SCREEN REPORTS, PER PLATE

$$\delta_{p,c} = E[x \mid \text{plate } p, \text{ compound } c] - E[x \mid \text{plate } p, \text{ DMSO}]$$

Treated minus control, on the same plate. Scaling fit on the field's DMSO wells only, never on the plate being judged.



Q1 · DOES IT AGREE? [E8]

**Less with other labs, for every compound.**  
Grey: a plate's  $\delta$  against the mean  $\delta$  of its own lab's other plates. Pink: against each other lab's mean, averaged. AMG900: 0.93 and 0.80.

Q2 · DOES IT TRACK THE CONTROLS? [E8]

**Inside labs, yes.**  
Spearman between a plate's control-well distance to the other labs and its disagreement with them: 0.20 to 0.53 by compound; pooled 0.30, shuffled 0.01.

Q3 · DOES IT MATTER? [E8]

**Effect size differs 14% to 79% between labs.**  
Effect size is  $|\delta|$ . Coefficient of variation of the lab means, by compound; inside one lab, plate to plate: 10% to 44%.

# Shown in fluorescence.

## Bet on brightfield.

Every number here is Cell Painting: five stains. The product reads brightfield: none. That gap is the bet, testable on public data.

### WHAT IS SHOWN

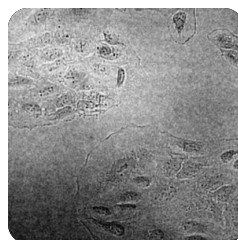
#### Stained wells name the lab

Untreated U2OS wells, 11 labs: the lab is named 99.9% of the time, chance 9%.  
results/e1

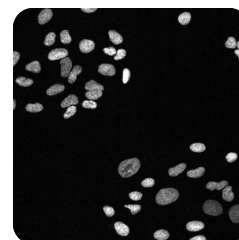
### WHY THE BET IS FAIR

#### Fewer variables, not more

No stain lot, dye lot or staining step:  
three unrecorded variables gone (A3).  
JUMP images brightfield beside the  
stains, so the same wells re-run without  
them.



BRIGHTFIELD



DNA STAIN

One field of  
one well,  
two ways.  
Lab 2, June  
2021.

### WHAT WOULD KILL IT

#### Brightfield alone at chance

Then the certificate asks for one cheap  
stain. The test: e1 on the brightfield  
planes of the same 93,228 wells.  
[pending]

Brightfield timecourse is the modality I build screens on.

# Would it cry wolf? 6 of 129.

A flag is only worth something if it is rare, and if it survives the choices made to compute it. Both are measured.

## LAB 2 · THE SAME 13 BATCHES, JUDGED TEN WAYS

|                                     | 7   | 14  | 21  | 12  | 19  | 26  | 2   | 8   | 16  | 23  | 30  | 20  | 3   |                |
|-------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|----------------|
|                                     | Jun | Jun | Jun | Jul | Jul | Jul | Aug | Aug | Aug | Aug | Aug | Sep | Oct |                |
| as on slide 06                      | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | 3,773 features |
| baseline is the first two batches   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | 3,773 features |
| baseline is the first three batches | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | 3,773 features |
| no clipping                         | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | 3,773 features |
| no image-level features             | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | 2,904 features |
| image-level features only           | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | 869 features   |
| all-vehicle plates removed          | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | 3,773 features |
| lab 1 removed from the field        | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | 3,773 features |
| lab 8 removed from the field        | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | 3,773 features |
| labs 1 and 8 both removed           | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | ■   | 3,773 features |

### HOW RARE

6 of 129 batches are outside, in 3 of 11 labs. 8 labs never flag.

### DOES EVERY PLATE AGREE

In lab 2's flagged batches 25 of 25 plates are outside. In its other 10 batches, 2 of 194.

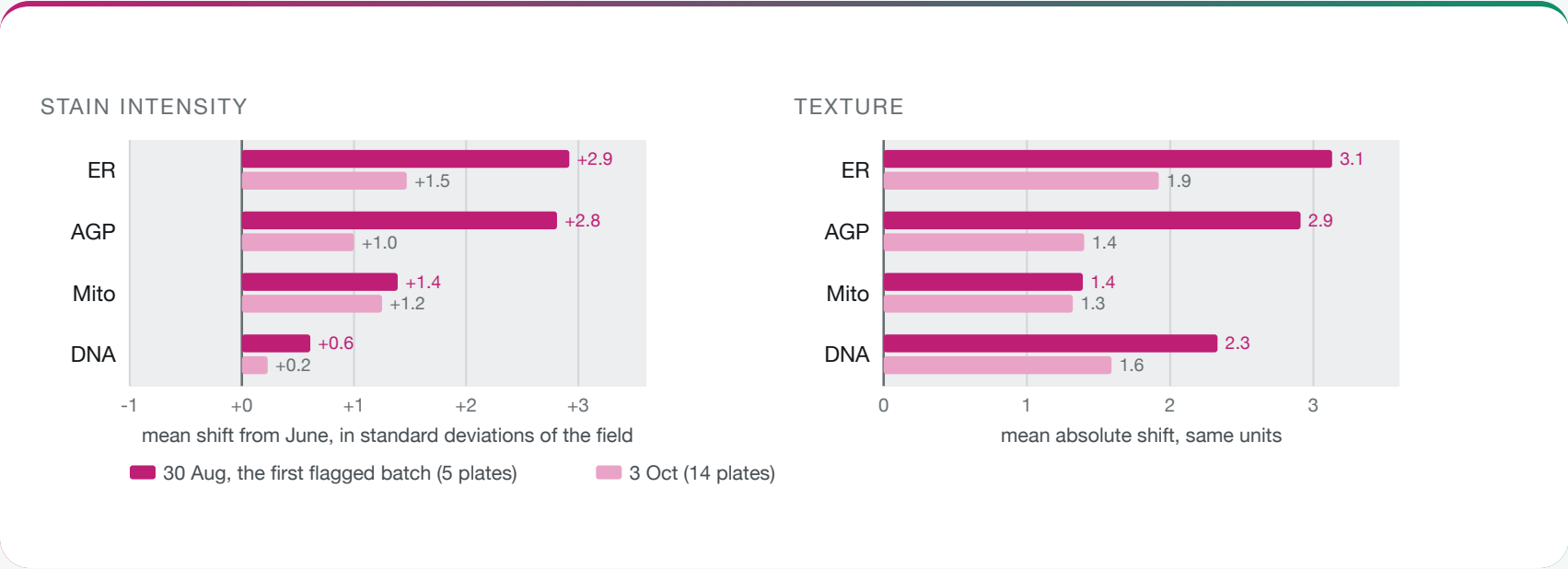
### WHAT CHANGES IT

1 of 10 variants: a baseline that takes in 21 June, the batch whose cell count fell. A baseline should be a batch the lab trusts.

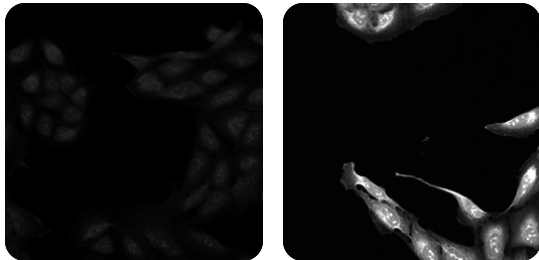
Outside: further from its own first batch than from the nearest other lab. Flagged batches hold 5, 6, 14 plates.

# Stain intensity rose. Cell count did not.

The certificate's distance is a difference of two means, so it reads feature by feature.



## ONE PIXEL WINDOW



7 JUN · ER      30 AUG · ER

7× brighter on the camera. Same lab, same line, same well type, twelve weeks apart.

Raw counts restored from each image's window. One field each.

**READING**

On 30 Aug, ER and AGP intensity sit 2.9 and 2.8 sd above June, and 96% or more of those features moved the same way. By 3 Oct: 1.5 and 1.0. Cells per well: 189, then 165 and 182.

**INSTRUMENT OR CELLS**

Image-level features carry 20% of the move and are 23% of the features, so the move is not confined to acquisition. Profiles cannot tell a stain lot from biology.

**WHAT A LAB GETS**

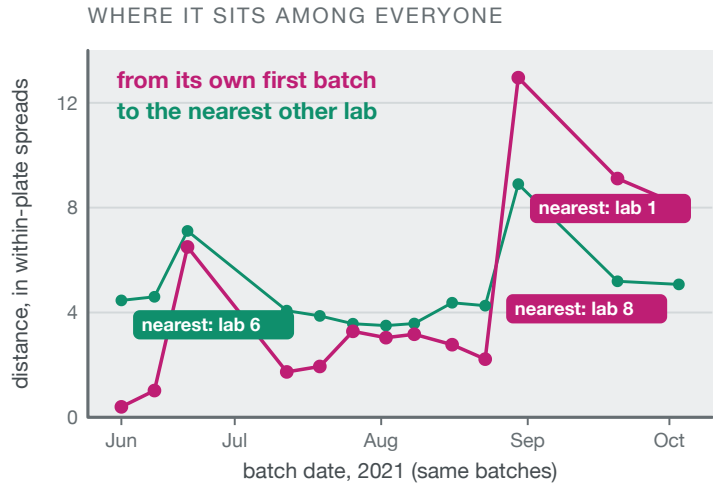
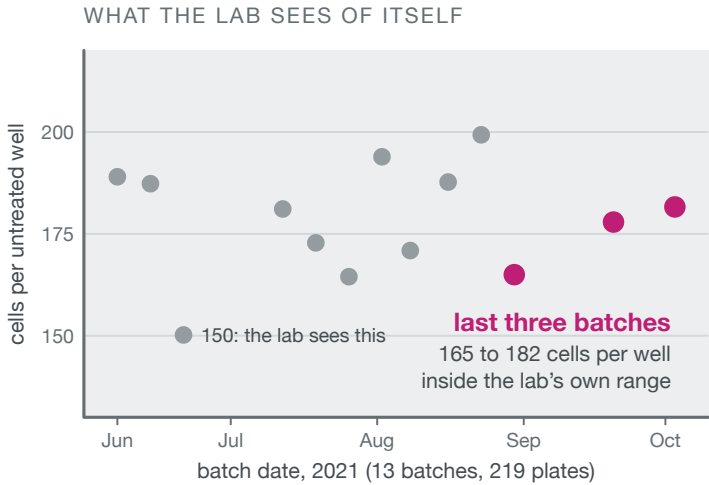
Where to look first: staining and exposure in two channels. Not why. The cause is the lab's to find, a day after the run and not four months.

Shift from the lab's first batch, in standard deviations of the field; means over the features of one channel.

# Alone, it looks fine.

## Among everyone, it moves.

Same 13 batches, two views.  
Left: the lab's own check. Right: its place on the map, and whom it resembles.



### LOOKING AT ITSELF

Cells per well stay between 150 and 199.  
Every batch passes. Odd things can sit comfortably inside a lab's own range.

### LOOKING AT THE MAP

From 30 Aug the batches sit further from the lab's own first batch than from another lab. The lab they most resemble changes: lab 6 → lab 1 → lab 8.

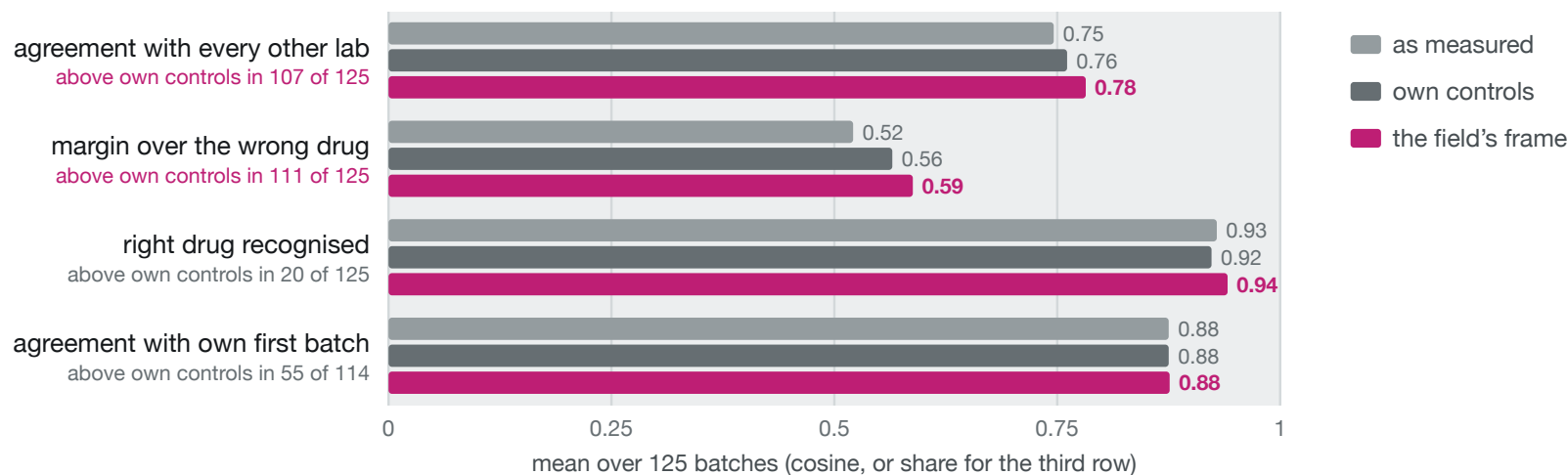
### WHAT A CHANGE MEANS

Not a verdict. Resembling a different lab than you did is a reason to look: interest when you meant to change something, concern when you did not.

Distances in units of the field's within-plate spread; the nearest lab is the lab whose centre is closest. Lab 2, U20S, 2021. JUMP [7]. What moved: A12.

# One frame: **what it buys.**

Each batch's untreated wells are moved onto the field's, built without that lab. The drugs are the test.



## DOES IT HOLD

With 100, 200 or 400 components and shrinkage from 0.05 to 0.5: above own controls in 89 to 116 of 125 batches for agreement, 52 to 123 for margin.

## WHAT IT DOES NOT DO

A lab's agreement with its own first batch does not change: 0.88, then 0.88. Effect sizes vary more between batches: 0.06 becomes 0.12. The gain is between labs, and it is modest: 0.76 to 0.78 over own controls.

## HOW DENSE

A lab's own baseline settles by eight plates: 0.89 to 0.39 spreads. The field's reference moves 0.41 when one lab leaves.

# Three buyers, one certificate.

Contributors use it free. The labs that cannot afford to be wrong pay.

|                  |  ACADEMIC LABS AND CORES |  PHARMA AND CRO SCREENING |  CELL BANKS AND VENDORS |
|------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| What is at stake | the paper                                                                                                   | the screen · ~\$2.6M                                                                                         | the lot                                                                                                    |
| What they pay    | free, for contributing controls                                                                             | ~\$5–15k per line, per quarter<br>170× less than one unchecked screen                                        | a fee per certified lot                                                                                    |
| What it builds   | the reference                                                                                               | the revenue                                                                                                  | the network                                                                                                |

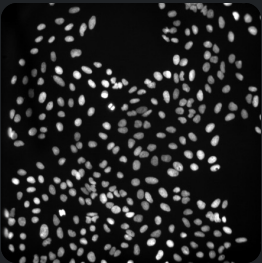
Proposed, not yet validated.

# Build the ugliest dataset in biology, **on purpose.**

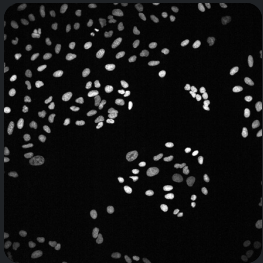
Same cells, same protocol, no drug. It already looks like this.

THE TEST SET · WHAT SIX LABS ALREADY SEND

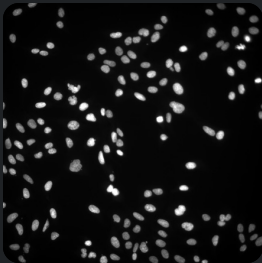
real wells, JUMP cpg0016, CC0



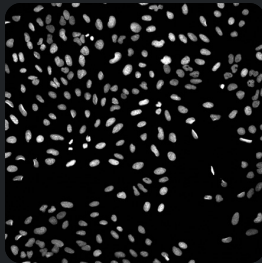
LAB 1  
Phenix · widefield



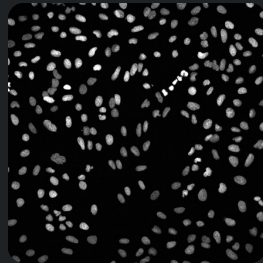
LAB 2  
CV8000 · confocal



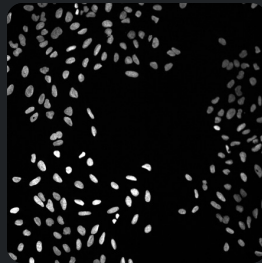
LAB 4  
Phenix · widefield



LAB 5  
CV8000 · confocal



LAB 6  
CV8000 · confocal



LAB 10  
CV8000 · confocal

**5**  
microscope models

**0.4–1.0**  
objective NA

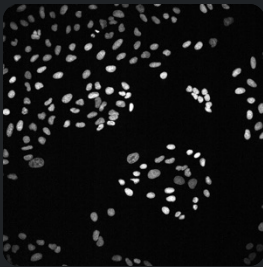
**0–3**  
brightfield planes

**0.60–0.70**  
µm per pixel

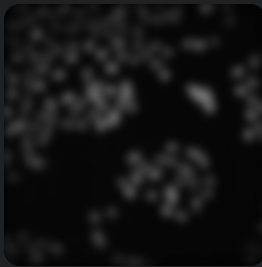
**8**  
plate types

Across the 13 JUMP labs.

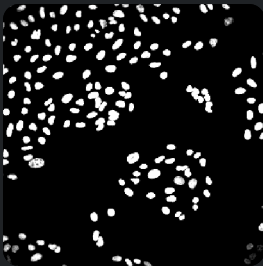
RED TEAM one well, simulated



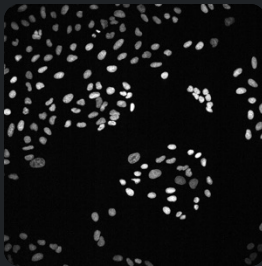
AS IMAGED



DEFOCUSED



OVER-EXPOSED



CLIPPED WELL

A public benchmark: name the failure.

The benchmark every method is graded on, normalization to bioactivity: the glue between the labs that make data and the companies that b

# A game, not a report card.

Every batch ranked on its untreated wells, never on results. Labs are a number unless they choose a name. The same arithmetic runs privately on your own plates with data access.

| LAB    | BATCHES | MEDIAN SCORE | BEST | STREAK | MEDIAN TYPICAL | MEDIAN TIGHT | MEDIAN STEADY | MEDIAN COUNT |
|--------|---------|--------------|------|--------|----------------|--------------|---------------|--------------|
| lab 5  | 33      | 70           | #1   | 33     | 3.35           | 0.27         | 0.81          | 0.31         |
| lab 6  | 13      | 60           | #16  | 13     | 4.44           | 0.41         | 1.23          | 0.12         |
| lab 10 | 17      | 54           | #10  | 17     | 3.84           | 0.62         | 1.21          | 0.23         |
| lab 8  | 4       | 52           | #54  | 4      | 4.10           | 1.08         | 0.86          | 0.09         |
| lab 11 | 5       | 47           | #23  | 2      | 3.79           | 0.87         | 1.57          | 0.17         |
| lab 3  | 13      | 44           | #18  | 6      | 3.27           | 0.80         | 2.01          | 0.29         |
| lab 7  | 8       | 42           | #69  | 8      | 6.38           | 0.58         | 1.17          | 0.33         |
| lab 2  | 13      | 39           | #48  | 10     | 5.68           | 0.57         | 1.99          | 0.14         |
| lab 1  | 6       | 32           | #78  | 6      | 8.68           | 1.10         | 2.03          | 0.09         |
| lab 9  | 11      | 25           | #97  | 11     | 9.97           | 0.50         | 1.76          | 0.45         |
| lab 15 | 6       | 23           | #68  | 6      | 5.53           | 0.87         | 1.69          | 0.44         |

### THE SCORE

Four measures in within-plate spreads, each a percentile among 129 batches: distance from the field’s centre, spread of the batch’s plates, distance from the previous batch, cells per well against the field. 100 minus the mean.

### THE BEST BATCH

lab 5, 28 Jun, 7 plates: score 82. Lab 5 holds 9 of the top ten and the longest streak, 33.

### YOUR CHOICE TO PLAY

Nothing in the score touches a treated well or a result, and nobody is named without opting in. The board is for climbing; the private version is for your own plates.

# Control wells leave. Nothing else does.

A control well has no compound, no target and no result, so there is nothing to clear. The map is built from those wells alone; the frame it produces is applied to your treated data by you.

## WHAT LEAVES THE BUILDING

### Untreated control wells only

images or the well-level profile of the wells with no compound. No treated well, no compound identity, no target, no result.

### A plate map of the control wells

which wells are controls; nothing about what the other wells hold.

### Nothing else

on this public replay that is 82,319 untreated wells from 10 other labs, and not one treated well.

## WHAT DATUM LINE DOES WITH IT

### Builds the map

one centroid per plate, in a space scaled on every other lab. The lab being placed never sets its own scale (3,773 features here).

### Answers

where the plate sits, distance from the lab's own baseline and to the nearest lab, cells per well, the verdict, the rank.

### Computes a move, not a model of you

the field's frame is one matrix per batch, fitted on your controls:  $x' = C_{\text{field}}^{\frac{1}{2}} C_{\text{batch}}^{-\frac{1}{2}} (x - m_{\text{batch}})$ , 200 components.

## WHAT IT NEVER SEES

### Your treated data

the move is sent to you and applied on your side. Your results are normalised in your building, by you.

### Your name, unless you choose

labs appear on the map and the board as a number by default; a lab opts in to a name. Every JUMP lab here is a number.

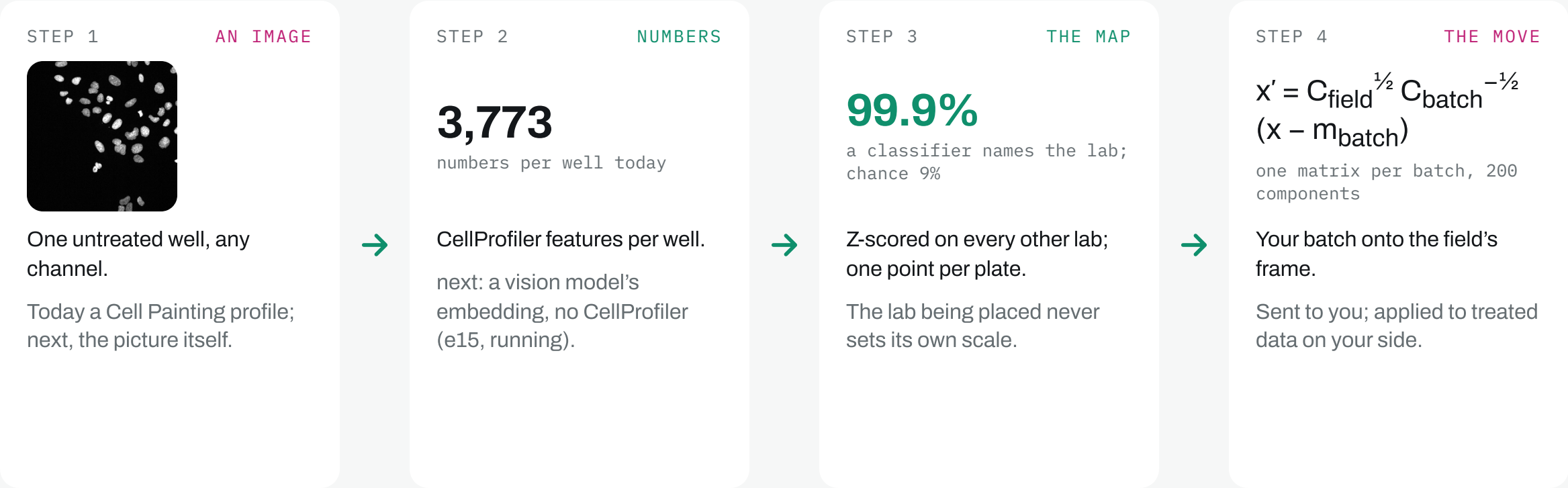
### Your images, for long

proposed: images are reduced to profiles on arrival and the originals deleted; the profile of a control well is what is kept.

Every analysis in this deck runs on this rule and is public: [github.com/kat-titter/datum-line/analysis](https://github.com/kat-titter/datum-line/analysis). The public map is JUMP, CC0 [7]. Retention and naming are proposed, not yet contracted.

# Image in. Place, distance, move out.

Four steps. The model is the cheap part; the map and the rule that nobody sets their own scale are the product.



PEOPLE  
the box, the glimpse, the game

AGENTS  
the API, the coordinates, the benchmark

BOTH  
every drop sharpens the map; every graded model raises the bar

One laptop runs the whole replay from cache in minutes. Every script: [github.com/kat-titter/datum-line/analysis](https://github.com/kat-titter/datum-line/analysis). What leaves a lab: A18.

# Cells cannot lie about where they came from.

Provenance, integrity and a tripwire, from the same map.  
Control wells only: nothing about agents, compounds or targets ever leaves a lab (A18).

## WHERE DID THIS COME FROM?

# 99.9%

of 93,228 untreated wells name their true lab, on plates the classifier never saw.

A plate claimed from one lab and made in another is caught by its cells. In the replay, 119 of 129 batches have every well naming its own lab (results/e1, e9).

## DO THE LABELS TELL THE TRUTH?

# 4 batches

65 plates whose plate map says positive control, and whose cells show no effect; no other well on those plates does either.

The data said what was done to the cells, not what the label said. Left out or mislabelled, the map cannot tell; that it is wrong, it can (results/e11).

## CELLS FROM NOWHERE

# 9 spreads

the farthest any of 129 real batches sits from every other lab. Beyond that, a culture is not the line it claims, or not alone.

Misidentified lines and contamination are the common case; untested here, and proposed as the first partner-lab test.

## SENTINEL CELLS

# proposed

An untreated well is a canary. If the controls move and nobody changed anything, something in the room did: a lot, a reagent, the air, the water.

Lab 2's autumn is what that looks like on public data (A13). Alerts across labs come after the certificate.

Defensive by construction: the map holds untreated wells, the benchmark grades methods on planted imaging failures, and no result or perturbation is stored. JUMP [7].

# About \$1.0M for eighteen months.

People first, then the benchmark. No lab of our own.

|                                  |                                                                                         |                |
|----------------------------------|-----------------------------------------------------------------------------------------|----------------|
| Founder                          | 18 months                                                                               | \$180k         |
| Technical co-founder             | 18 months                                                                               | \$180k         |
| ML engineer                      | from month 7                                                                            | \$160k         |
| Payroll overhead                 | 20% of salaries                                                                         | \$104k         |
| The benchmark dataset            | about 60 plates with planted failures, run where the cells already are; a CRO if needed | \$120k         |
| Partner-lab bench time           | three labs image the controls they already take; reagent top-ups and shipping           | \$30k          |
| Compute and storage              | images, models, hosting, the API                                                        | \$45k          |
| Legal, IP, insurance, accounting | incorporation to first contract                                                         | \$45k          |
| Travel and partner visits        | two benchmark partners, three labs                                                      | \$20k          |
| Contingency                      | 15%                                                                                     | \$133k         |
| <b>Eighteen months</b>           | about                                                                                   | <b>\$1.02M</b> |

NO IN-HOUSE LAB

Partner labs image the controls they already take. The benchmark runs where the cells already are, or at a CRO. A lab of our own would cost more than the benchmark and prove less.

THE UGLY DATASET IS INSIDE

\$150k of \$1.02M, about 15%, builds the benchmark and the bench time around it. It is milestone three, not a later round.

WHAT IT HAS TO SHOW

Three labs live on the map, a model that reads any image, the benchmark with two partners, and a first paid certificate. Then a seed.

Salaries are below market and assume founders are paid. Nothing here is committed; the proof of concept sets the size.